

AL REVIEW

Exam

- 6 Questions

- 98 marks available. 99 if you read instructions

1Q - First Iso Thm

1Q - Thm & proof (one we've seen) fill in the blanks (Understand stuff about homomorphisms, etc. etc.)

1Q - P. groups (brings in 3rd Iso) (Maybe most difficult)

1Q - Sylow - Thm to prove g not simple $\rightarrow$ look at example 4.15
 $\rightarrow$ be ok with that

2Q's - Rings $\rightarrow$ all Q's have appeared somewhere

$\hookrightarrow$ Q's order matters (not necessarily commutative)

Be able to prove things are subrings

Quite a bit of give defⁿ & give Thm.

check well-defined (g.p. actions, homomorphisms)

$$\tau(g'H) = \tau(gH) \\ \text{if } g'H = gH$$

Lagrange's Thm

G is finite & $n \leq 6$, then $|H| \mid |G|$

Cauchy's Thm

If $p \mid |G|$ then G has an element/subgroup of order p

Thm 5

Q4

Correspondence Thm

$$\begin{aligned} \text{a) Since } K \trianglelefteq G &\Rightarrow K/H \trianglelefteq G/H & B \trianglelefteq A \\ L \trianglelefteq G &\Rightarrow L/H \trianglelefteq G/H & C \trianglelefteq A \\ K \trianglelefteq L &\Rightarrow K/H \trianglelefteq L/H & B \trianglelefteq C \end{aligned}$$

$$\text{b) By third iso } (A/K)/(C/B) \cong (A/C) \quad (A/C) = G/H / L/H \cong G/L$$

Tut 12

Q6

a) integral domain
↳ no zero divisors
↳ commutative

i) Suppose Neither R or S is an integral domain.

There exists $r, r' \in R$, $s, s' \in S$ with $r, r' \neq 0$, $s, s' \neq 0$ such that $rr' = 0_R$ and $ss' = 0_S$

Consider $(1, s), (r', s') \in R \otimes S$

$$(1, s)(r', s') = (r', ss') = (0_R, 0_S) = 0_{R \otimes S} \quad \text{Thus } R \otimes S \text{ has a zero divisor a contradiction}$$

$$\text{Since } (1, s), (r', s') \neq (0_R, 0_S)$$

ii) Let $s \in S$

Consider $(0_R, s), (1, 0) \in R \otimes S$

$$\text{Then } (1, 0)(0, s) = (0, 0)$$

Since $R \otimes S$ is ID

One of $(1, 0)$ and $(0, s)$ must be $(0_R, 0_S)$

$$\text{Thus } (0, s) = (0_R, 0_S) \quad \text{so, } s = 0$$

b) Skip

c) $R = \mathbb{R}$, $S = \mathbb{R}$

Consider $(3,0)$ for all $(a,b) \in (\mathbb{R} \times \mathbb{R})$ $(3,0)(a,b) = (3a,0)$ $\therefore (3,0)$ is not a unit, thus $\mathbb{R} \times \mathbb{R}$ is not a field.

Q1

a) $x = r+r$ and $x^2 = r+r$

$$(r+r)^2 = (r+r)$$

LHS: $(r+r)^2 = (r+r)(r+r)$ } distributivity

$$= r^2 + r^2 + r^2 + r^2$$

$\uparrow$
 $r^2 = r+r$

$$= r^2 + r + r + r + r^2$$

Thus $r+r = r^2 + r + r + r^2$

$$0 = r^2 + r^2$$

So, $0 = r+r$

b) $r+r=0$ by a, so $r=-r$

c) We need to show $\forall r,s \in R$ $rs = sr$

Let's consider $x = r+s$

$$(r+s)^2 = r+s$$

$$(r+s)(r+s) = r^2 + rs + sr + s^2$$

Thus $r^2 + rs + sr + s^2 = r+s$

$$r^2 + rs + sr + s^2 = r^2 + s^2$$

$$rs = -sr$$

$$rs = sr$$

$$rs = sr$$